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Broadband RF Design with MEEP: A Set of 3D-Printed Open-Source RF Horns in the multi-GHz Bandwidth

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Abstract

Radio-frequency (RF) antenna design and fabrication traditionally proceeds in three phases. First, the design performance is modeled with expensive, proprietary software. Second, the design is built using precision metal machining or printed circuit board (PCB) fabrication. Third, the result is characterized using RF measurement tools. The process can include machine learning for optimization. We have developed an open-source alternative process that utilizes the MIT Electromagnetic Equation Propagator (MEEP) package for design and simulation, and open-source 3D printing with conductive polymer for fabrication. Using this process, we created a set of broadband RF horn antennas. The radiation patterns and the VSWR match our MEEP calculations, while the cross-polarization ratio matches theoretical expectations. These results indicate that our creations are linearly polarized, broadband RF horn antennas with bandwidth [4.0-18.0] GHz, or a fractional bandwidth of 127%. Because the maximum frequency of our network analyzer is 18 GHz, these results are lower limits. Our results therefore represent a successful broadband open-source CEM/CAD workflow utilizing direct, repeatable conductive printing that circumvents the need for electroplating processes.

Keywords: Computational Electromagnetism (CEM), Additive Manufacturing, MEEP, RF Engineering, Open-Source Design

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1. Introduction

Broadband RF antennas are ubiquitous tools within scientific instrumentation and communications. Traditionally, these are designed with expensive, proprietary software packages [1,2]. Designs are fabricated by cutting metal with precision machine tools (subtractive manufacturing). Compared to additive manufacturing techniques, subtractive manufacturing can be costly for rapid prototyping [3,4]. Open-source additive manufacturing boosts cost efficiency in both the design and fabrication, provided the 3D printing filament has sufficient conductivity.

Development of new RF antenna designs requires exploration of conductor shapes that set the boundary conditions for radiated signals. Developing new shapes that meet design requirements has been aided by machine learning (ML) strategies [5–7]. While mixing new ML algorithms into open CEM code is straightforward, incorporating them into proprietary software is often challenging. Ideally, RF engineers would keep the functionality of proprietary software, while retaining the flexibility to incorporate new ML techniques. Creating a design process based on open CEM tools achieves this goal.

Previously, J. C. Hanson (2021) demonstrated that the open-source MEEP software package may be used as an RF antenna design tool [8]. MEEP operates via the FDTD algorithm for Maxwell's Equations on a Yee lattice [9]. The key to the technique is that MEEP is written to account for the scale-invariance of Maxwell's Equations. MEEP is normally used for designs with wavelengths at the μm scale. Designs with cm wavelengths may be created by reinterpreting length units in MEEP. Using this approach, J. C. Hanson produced RF phased array designs in two and three dimensions using MEEP functionality [8]. The designs included broadband RF horn elements in the [0.5-5] GHz bandwidth. Further, J. C. Hanson demonstrated that open-source CAD software may be used to create complex designs that can be 3D printed [10].

In recent years, RF antenna systems have begun to be fabricated with 3D printing. The authors of [11] presented a hybrid strategy that combined traditional PCB fabrication, non-conductive 3D printing, and metallization of 3D printed structures. The assembled structure was a low-cost, electronically steerable MIMO array for mm-Wave 5G applications. The authors of [12] produced a waveguide-fed holographic beam-forming antenna using a dual print-head 3D printing system that combined non-conductive PLA supports with conductive filament. The result was a steerable beam at frequencies near 10 GHz. The authors of [13] produced a Mills-Cross, frequency-diverse, metasurface antenna capable of electronic steering in the K-band, using a mixture of non-conductive PLA and conductive filament. Finally, the authors of [14] printed a holographic metasurface antenna using the same technique found in [13] for Fresnel-zone focusing on and off-axis in the X-band.

In this work, we present the first open-source, broadband, 3D printed set of RF horns, designed with MEEP and fabricated entirely from conductive 3D-printer filament. Note that the authors are MEEP users, but not MEEP developers. Where appropriate, we have cited the creators of MEEP [9]. In Sec. 2, we explain the parametric design, CAD, and fabrication techniques. In Sec. 3, we show that the **E**-plane and **H**-plane radiation patterns, the VSWR, and the cross-polarization ratios match predictions. In Sec. 4, we discuss the limitations of our analysis, summarize key results, and discuss applications and future directions for this research.

2. Methods and Materials

The goal of this work was to use open CEM tools to design, model, and fabricate linearly polarized RF horn antennas in the multi-GHz range via additive manufacturing. Figure 1 contains a diagram of the design and fabrication process. Using parametric CAD design, we can specify antenna shapes using analytic functions that are then translated into the open CAD design tool kLayout. Antenna designs created with kLayout can be imported into MEEP via the GDSII file format. GDSII files can also be translated into 3D printer files, ensuring that the device and model match. In Fig. 2 (left), the CAD design in kLayout is shown. In Fig. 2 (middle) and (right), the model is rendered using Mayavi.

Our open fabrication technique hinges on the existence of commercially available conductive 3D printer filament. Multi3D[1] has shown that a copper-doped thermoplastic serves as a 3D printer filament with resistivities of $\approx 10^{-2}\ \Omega$ cm, depending on the printing pattern. Using a tungsten-tipped 3D printing head and Electrifi filament from Multi3D, we can complete antenna prints with volumes of ≈ 350 cm^3 in several hours. The printed structures are fitted with RF connectors and tested as receivers on custom mounts that swivel in zenith and azimuthal angles. A network analyzer with bandwidth [0-18] GHz was used to measure complex S11 and S21 parameters, with a calibrated log-periodic dipole array (LPDA) antenna as the transmitter.

[1] See https://www.multi3dmaterials.com

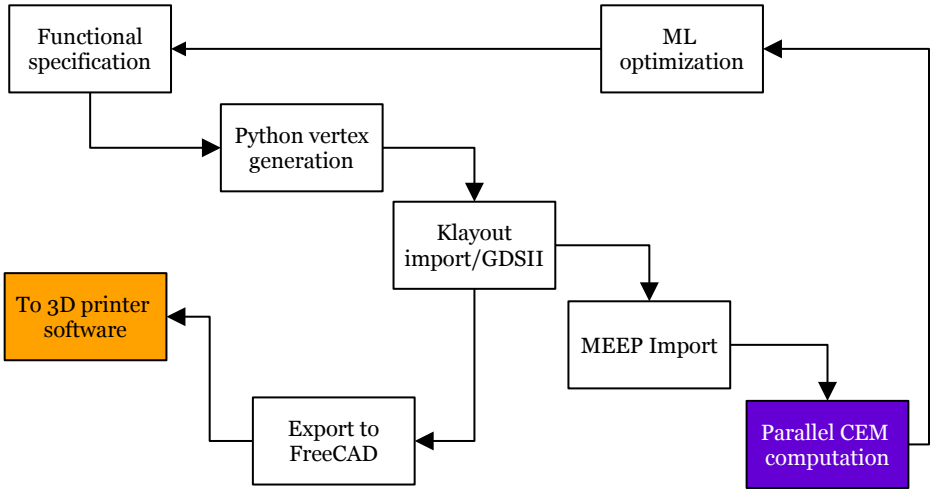


Figure 1. Flow diagram for the open-source design and fabrication process.

Table 1. Design parameters for the RF horn antenna.

Parameter	Variable Name	Value [cm]
Cavity Length	c_l	0.77
Cavity Width	c_w	1.00
Surface Length	s_l	16.52
Opening Width	w	9.59
Height	z	2.5

2.1. CEM Design and Open-Source CAD

The open-source CEM design begins with choosing parametric design parameters for the exponential RF horn and importing them into kLayout (Fig. 1-2). Let the origin refer to the center of the left edge of the cavity in Fig. 2 (left). The horn cavity is the rectangular volume where the coaxial cable attaches, with xy-dimensions $c_l \times c_w$ (Tab. 1). The horn surfaces are the curved structures that connect the cavity to open space, with overall length s_l in the x-direction, including the cavity. The opening is the area where the surfaces stop and radiation exits the horn. The opening has width w and height z in the y and z-directions, respectively. Using these variables, Equations 1 and 2 specify the shape of the RF horn surfaces in the xy-plane:

$$f(x) = \frac{c_w}{2} \exp(k(x - c_l)) \tag{1}$$

$$k = (s_l - c_l)^{-1} \ln\left(\frac{w}{c_w}\right) \tag{2}$$

The function $f(x)$ in Equation 1 describes the upper surface in Fig. 2 (left), while $-f(x)$ describes the lower surface. Table 1 contains the values for c_l, c_w, s_l, and w. The RF horn is designed to be linearly polarized in the y-direction. The RF horn design has a height z, out of the page in Fig. 2 (left), and each conductive and dielectric surface, including the coaxial cable, has a specified thickness in the xy-plane and z-direction.

In previous studies [8], antenna structures were instantiated directly as native MEEP objects. In the present workflow, the antenna geometry is instead generated parametrically and transferred to kLayout, where it can be inspected and exported in GDSII format. The same geometry is then used for the MEEP calculation and for conversion to a printable 3D model. This approach preserves the flexibility of a scripted parametric design while

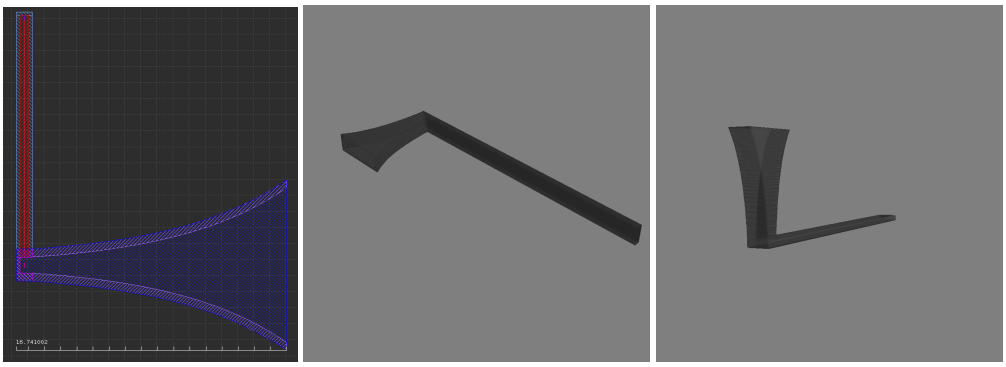


Figure 2. (Left) The kLayout CAD design for the RF horn (top view), with a coaxial cable attached. The x-direction is to the right, the y-direction is up, and the z-direction is out of the page. The white ruler gives 16.741 cm. (Middle and Right) The RF horn and cable model from alternate angles.

avoiding the need to reproduce a complex antenna from many simple MEEP primitives. The implementation code used to generate and simulate the antenna is maintained in the open-source project repository cited in the Data Availability Statement.

2.1.1. VSWR Calculations

The voltage standing wave ratio (VSWR) is expressed in terms of the magnitude of the complex reflection coefficient Γ at the antenna input port [15]:

$$\mathrm{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{3}$$

In the MEEP model, a broadband excitation is launched through the modeled coaxial feed. A MEEP `FluxMonitor` is used to separate the incident and reflected contributions at the input line, from which Γ and the corresponding VSWR are calculated[2]. The model is fully three-dimensional and uses the material properties of the Electrifi filament. Implementation details and source code are provided in the open-source repository cited in the Data Availability Statement.

2.1.2. Radiation Pattern Calculations

We compute the antenna radiation pattern in MEEP using near-to-far field projection[3]. A `NearToFarRegion` encloses the radiating structure, and the `meep.get_farfield()` routine is used to evaluate the complex electric and magnetic fields in the far field. In the coordinate system of Fig. 2, the horn boresight is the $+x$ direction, the xy-plane is the **E**-plane, and the xz-plane is the **H**-plane.

The time-averaged Poynting vector is

$$\mathbf{S} = \frac{1}{2}\,\mathrm{Re}(\mathbf{E} \times \mathbf{H}^*), \tag{4}$$

and the radiated power used for the angular pattern is the component projected along the radial direction,

$$P_\mathrm{r} = \mathbf{S} \cdot \hat{\mathbf{r}}. \tag{5}$$

[2] See https://meep.readthedocs.io/en/latest/Python_Tutorials/GDS_Import.
[3] See https://meep.readthedocs.io/en/master/Python_Tutorials/Near_to_Far_Field_Spectra.

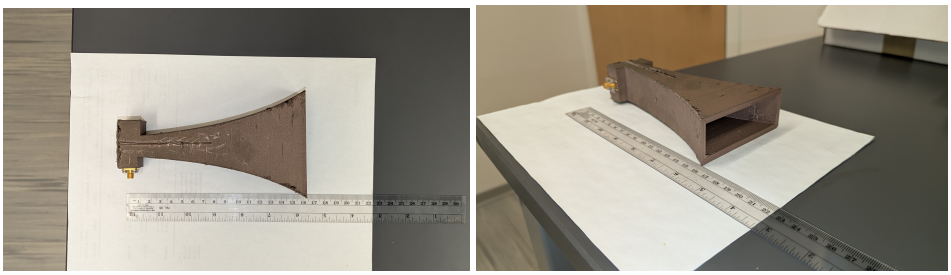


Figure 3. Views of an example of our 3D printed RF horns. The horn is approximately 16 cm long.

For the **E**-plane, $\hat{\mathbf{r}} = (\cos\phi, \sin\phi, 0)$, while for the **H**-plane, $\hat{\mathbf{r}} = (\sin\theta, 0, \cos\theta)$. The normalized radiation pattern is reported in decibels as

$$P_{\text{norm}}(\alpha) = 10\log_{10}\left[\frac{P_{\text{r}}(\alpha)}{\max(P_{\text{r}})}\right]. \tag{6}$$

The corresponding source code is maintained in the open-source repository cited in the Data Availability Statement.

2.2. Fabrication Technique

The parametric horn geometry was converted from the kLayout/GDSII design into a printable 3D model and fine-tuned for fabrication. The antennas were printed directly from Multi3D Electrifi conductive filament on a Prusa XL equipped with a tungsten-tipped nozzle. Electrifi was printed at approximately 135–145°C without bed heating. Because the horn is tall and narrow, masking tape with adhesive and removable buttresses were used to improve build-plate adhesion and mechanical stability during printing. After printing, the temporary supports were removed and an SMA connector was attached to the cavity so that the center conductor extended into the feed region. No secondary electroplating or conductive coating was applied to the radiating structure. One example of the RF horns is shown in Fig. 3, with a 12-inch ruler for scale.

3. Results

Using the techniques described in Sec. 2.2, we fabricated eight RF horns with the parameters in Tab. 1. As listed in Tab. 1, the RF horn is 16.5 cm long (see ruler in Fig. 3). The dimensions of the opening are 2.5 x 9.59 cm, including the 0.25 cm thickness in the z-direction. The SMA connector is mounted to the cavity, with a coaxial pin extending into the chamber such that the radiation is linearly polarized in the y-direction (see Sec. 2.1). A network analyzer was used to measure the S11 and S21 parameters across the bandwidth. In Sec. 3.1, the S11 results are formed into the VSWR parameter, and graphed versus frequency. In Sec. 3.2, the S21 results are formed into radiation patterns, and plotted versus the azimuth and zenith angles.

3.1. VSWR Results

The VSWR results are presented in Fig. 4. The 3D printed RF horn was mounted to a tripod and pointed at the ceiling. The antennas were connected to port 1 of an Agilent Technologies E5063A network analyzer [100 kHz - 18 GHz] via a 6 ft. Coleman Cable 8421F M17/75-RG214 N-type coaxial cable and an SMA to N connector. The network analyzer was calibrated across the entire bandwidth using a calibration kit consisting of short, open, and 50 Ω terminators. Once calibrated, the network analyzer was used to record the S11 parameter for each of the eight RF horns. The analyzer was set to average mode with 999 traces in the average (2.5% smoothing), using the entire bandwidth. The trace was recorded for each horn, and the data in Fig. 4 represent the average across all horns.

Our MEEP/CAD matches the dimensions of the RF horns. We assume a resistivity of 0.01 Ω cm, from the Electrifi technical specifications (https://www.multi3dmaterials.com). The simulation is first executed assuming only the coaxial cable portion. The dielectric portion of the coaxial cable was modeled in MEEP with relative permittivity $\varepsilon_r = 2.0$ and a loss tangent of $\approx 2 \times 10^{-2}$ at 4 GHz. All other surfaces, including the exterior of the cable, were modeled using `mp.metal` (ideal conductivity). An RF pulse is sent from a source coaxial with the cable conductor. The pulse energy passes a `FluxMonitor` MEEP object, recording the flux at each frequency bin. The simulation is then reset, and the RF horn structure is added to the cable. The simulation is run a second time, with -1 times original flux pre-loaded to the `FluxMonitor`. The reflection coefficient Γ was calculated by taking the square-root of the complex ratio of reflected flux (minus the incident flux in the second run), to the incident flux in the first run. The VSWR is found using Eq. 3.

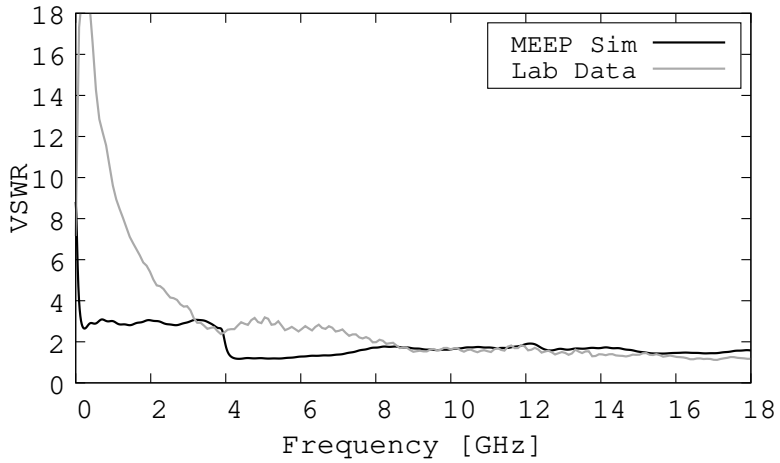


Figure 4. Average VSWR for all antennas (gray line), and MEEP calculation of the VSWR (black line), assuming `D_conductivity=0.02`.

An exponential decrease in VSWR between 0 and 4 GHz is evident in the lab data. The VSWR data reach $\lesssim 2.5$ at 4 GHz, when the free-space $\lambda/2 \approx 3.75$ cm. The data therefore indicate impedance matching between the coaxial cable and RF horn above 4 GHz. The VSWR approaches 1 with increasing frequency, and shows no sign of rising even as the 18 GHz bandwidth limit is reached. Thus, we conclude that the conservative estimate on the measured bandwidth is [4-18] GHz, with a fractional bandwidth of 127%. The theoretical maximum fractional bandwidth is 200%, and the 127% value is limited by the instrumentation, not the 3D printed antennas.

Above 4 GHz, the MEEP calculations accurately predict the VSWR to be between 1 and 2. Below 4 GHz, there are two effects that alter the simulation relative to the lab data. First, 4 GHz corresponds to an unrealistic mode cutoff in our cable model. The coaxial cable model is a rectangular prism from our CAD with the correct lateral dimensions, extended into the z-direction. In our GDSII file, there is a central conductor, a surrounding dielectric layer, and outer conductor layers. This works well above 4 GHz, where mode cutoffs are irrelevant. Second, the finite conductivity of the dielectric must be chosen such that the loss tangent gives the right behavior at lower frequencies. We find that loss tangents corresponding to the MEEP parameter `D_conductivity` of 0.02 improve the match below 4 GHz. The smooth transition from high to low VSWR, however, is not modeled.

3.2. Radiation Patterns

The radiation pattern results are presented in Figs. 5 and 6. Again, the 3D printed RF horn was mounted to a tripod. For these measurements, the 3D printed horns were

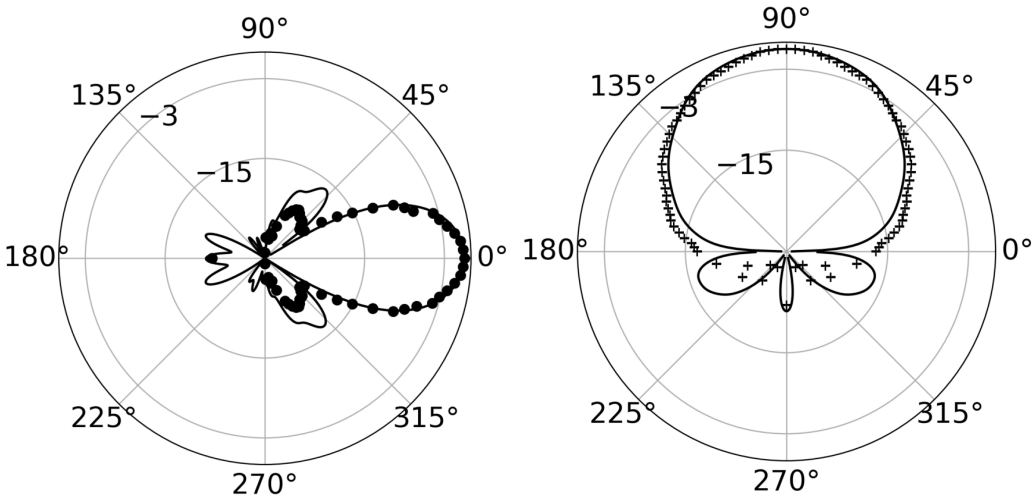


Figure 5. (Left) E-plane radiation pattern data (black circles) for HORN1, at 5.9 GHz, along with the MEEP calculation (line). (Right) H-plane radiation pattern data (black crosses) for HORN1, at 5.9 GHz, along with the MEEP calculation (line).

connected to port 2 of the network analyzer via the RG214 N-type coaxial cable and an SMA to N connector. Port 2 was configured as the receiving port. Port 1 was configured as the transmitting port, and connected to an A.H. Systems SAS-510-7 log-periodic dipole array (LPDA) via a Gore PhaseFlex coaxial cable. The LPDA and 3D printed horn were co-polarized, and mounted to a separate tripod. The network analyzer was calibrated and set to the bandwidth [5.5-18] GHz, where the received S21 signal was $>$ 10 dB above thermal backgrounds. No RF amplifier was used in the S21 setup. The S21 parameter was recorded at 2.5 degree intervals in the E-plane and H-plane for HORN1. The different angles were arranged by adjusting the tripod mounting mechanisms, and measured with an electronic level. Once the technique was established, the S21 was measured at 10 degree intervals for the E-plane and H-plane for HORN1-HORN8. For each angle and horn, the recorded S21 trace was an average of 999 measured traces (2.5% smoothing).

Figure 5 contains the data taken for HORN1. Figure 5 (left) corresponds to the E-plane at 5.9 GHz, and Fig. 5 (right) corresponds to the H-plane at 5.9 GHz. At the time HORN1 was printed, we had access to an older network analyzer with [100 kHz - 6 GHz] bandwidth. Thus, S21 was averaged from 5.8 to 6.0 GHz. For the E-plane graph, the angle being plotted is the azimuthal angle in the xy-plane, and the boresight corresponds to 0 degrees. For the H-plane graphs, the angle being plotted is the zenith angle in the xz-plane, and the boresight corresponds to 90 degrees. Normally, the S21 power is disaggregated by frequency, and separate radiation patterns are graphed for separate frequencies. However, as we averaged over [5.8-6.0] GHz, there is a single radiation pattern at the center frequency of 5.9 GHz. Radiation patterns are typically normalized to the S21 power at boresight, so the largest power is 0 dB in each plot.

The MEEP calculation matches the main lobe precisely in both cases, E and H-plane. For the MEEP model, it was sufficient to treat the Electrifi material with `mp.metal` (ideal conductivity). Treating it as a material with finite conductivity made little difference in the results. MEEP correctly predicts that the width of the pattern is narrower in the E-plane compared to the H-plane. Side lobes are visible in the patterns at $\pm$45 degrees in the E-plane. Side lobes with relative powers of -15 dB are common in RF horn designs. MEEP correctly predicts the location of the side lobes, and comes close to modeling the side-lobe-level. MEEP also correctly characterizes the back lobe levels in the E and H-plane, which are also below -15 dB. Side lobes and back lobes are caused by radiation that diffracts around the edge of the RF horn opening.

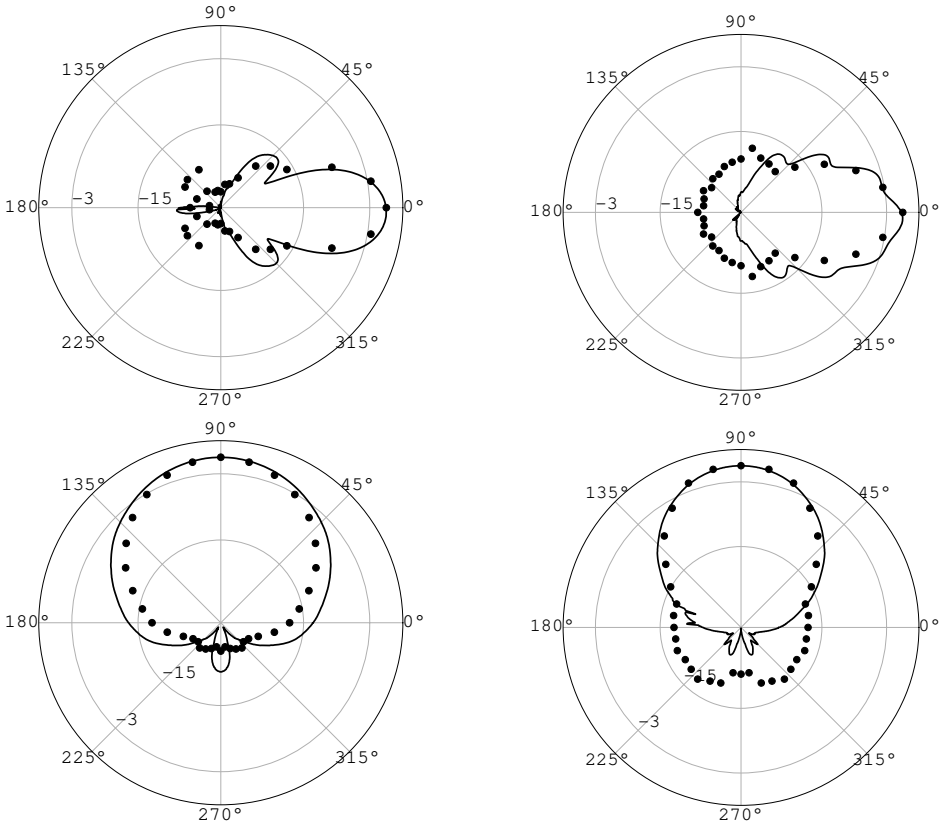


Figure 6. (Top left) Average E-plane data (black dots) for HORN1-HORN8, at 6.3 GHz, and MEEP model (black line). (Top right) Same, at 12.3 GHz. (Bottom left) Average H-plane data (black dots) for HORN1-HORN8, at 6.3 GHz, and MEEP model (black line). (Bottom right) Same, at 12.3 GHz.

Figure 6 contains the average data taken from HORN1-HORN8. The S21 power was disaggregated by frequency, and separate radiation patterns are graphed for separate frequencies, but averaged across RF horns. The frequencies were binned in 1 GHz bins, and the first bin center is at 6.3 GHz, so the lower edge of that bin is 5.8 GHz. Below this frequency, noise started to interfere with our S21 measurements. Thus, the frequency bins centers were set to 6.3, 7.3, ..., and 17.3 GHz. The average E-plane radiation pattern at 6.3 GHz is shown in Fig. 6 (top left), while (top right) shows the same at 12.3 GHz. The H-plane data at 6.3 GHz is shown in Fig. 6 (bottom left), while (bottom right) shows the same at 12.3 GHz. The same conventions from Fig. 5 are used in Fig. 6.

The MEEP calculation matches the main lobes, and predicts some side lobe and back lobe behavior. The match becomes more challenging at high frequencies when higher-order modes begin to fit within the horn body. At low frequencies, the side-lobe-levels and back lobe levels and locations are well modeled. At high frequencies, we measured larger back lobes than those predicted by MEEP. However, this is likely due to leaked power in our measured data. At high frequencies, we observe power captured by the N-to-SMA connector and cable, even without the 3D printed antennas attached. This leaked power level is small, but measurable at the -15 dB level. Apparently, it does not occur in every SMA-RF horn connection, because it was not observed in the data in Fig. 5.

An interesting facet of RF antenna fabrication arose after 3D printing our units. The raw units finished with spurs and droplets of extra filament that acted as miniature antennas. These surface imperfections decreased the initial S21 power. After removing them with files and sandpaper, the inner surfaces of the horns were smoother. The results for HORN4 are shown in Fig. 7. The data in Fig. 7 are binned in 1 GHz bins, centered at 6, 7, ..., and 18 GHz. The result was an increase in S21 power that varies between 2 and 10 dB, and

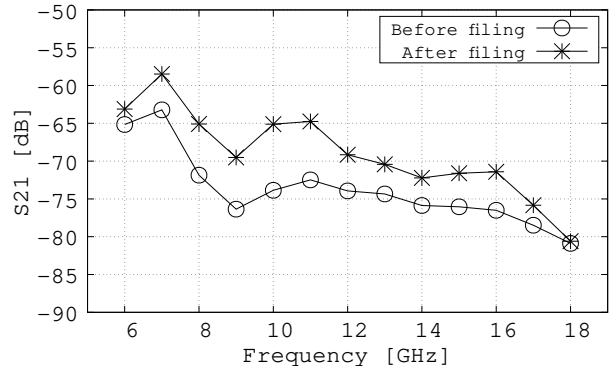


Figure 7. The S21 versus frequency before (circles) and after (stars) filing the 3D printed horn.

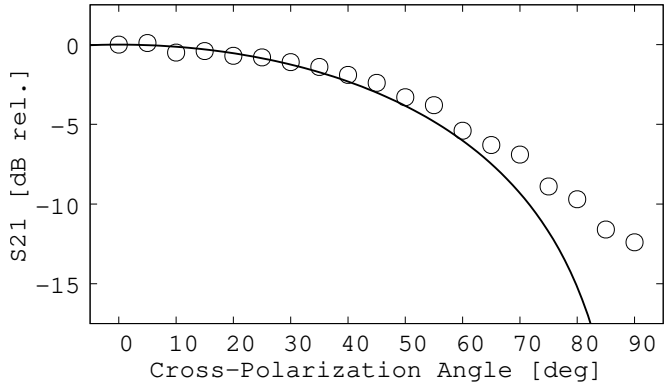


Figure 8. (Circles) The measured cross-polarization ratio in dB. (Line) The prediction from Eq. 7.

the largest gain is found near 10 GHz. It is difficult to explain the subtle variation in S21 gain due to smoothing and filing, because the imperfections varied in size. However, we learned that smoothing is an important step in the fabrication process.

3.3. Cross-Polarization Ratios

The cross-polarization results are shown in Fig. 8. The cross-polarization ratio was measured with the 3D printed RF horns and LPDA arranged in the same S21 configuration as the radiation pattern data. The cross-polarization angle was measured with an electronic level. The prediction for the dependence of received power versus cross-polarization angle is known from optics and antenna theory. Let P_r be the received power, P_t be the transmitted power, and θ be the cross-polarization angle. These are related by

$$P_r = P_t \cos^2 \theta \tag{7}$$

When normalized, the ratio P_r/P_t in dB is equivalent to normalized S21. The results for HORN1 at 5.8 GHz are shown in Fig. 8, and show good agreement with Eq. 7. As the cross-polarization angle approaches 90 degrees, the prediction from Eq. 7 tends to negative infinity in dB. In this regime we find a cross-polarization ratio of about -12 dB, which is comparable to the measured side-lobe-level ($\lesssim -15$ dB). A good reference for Eq. 7 may be found in the section on polarization mismatch in [16].

4. Conclusions and Next Steps

This work establishes an open-source design and fabrication process joining parametric design, open-source CAD, CEM with MEEP, and conductive 3D printing via Electrifi filament and Prusa XL printers. Our process eliminates the need for electroplating or coating, and the need for expensive, proprietary modeling software. The VSWR, radiation

pattern, and cross-polarization data consistently match expectations and, in most cases, are well-modeled by MEEP. Our fabricated designs have measured bandwidth of at least [4-18] GHz, with fractional bandwidth $> 127\%$.

The next step in the research is to assemble the RF horns into a phased array. Using a 1-to-8 RF power splitter, and programmable phase shifters, we plan to show that the collective beam from 1×8 and 2×4 configurations is steerable and broadband. Thus, we have taken a significant step towards producing the first completely 3D printed, RF broadband phased array. Once we have shown that the phased array is steerable, we plan to investigate the broadband capabilities of the design. For example, broadband phased arrays can produce grating lobes that can be steered independently by sweeping the frequency across a wide range. Coincidence between a low-frequency beam steered with phase shifts, and a high-frequency beam steered with a sweeping frequency, could represent an interesting way to improve sensing capability.

Author Contributions: J. C. Hanson conceived of the open-source design process, produced the MEEP calculations that led to the first RF horn design, identified the conductive 3D printer filament, incorporated open-source CAD, and produced fully three-dimensional CEM models to compute the VSWR and radiation patterns for the fabricated designs. A. Wildanger and Gia Bao Dong sourced materials, and imported CAD designs into 3D printing format, and completed 3D prints to produce the designs. S. Smith, A. Song, and N. Waldorf wrote and tested MEEP models, greatly improving code organization and structure, while exploring new ways to steer phased arrays. A. Waseem installed and validated `gpmeep`, a GPU version of MEEP, on our HPC. This accelerated the MEEP computations. A. Waseem also made corrections to the RF coaxial cable model for the VSWR calculation. Finally, J.C. Hanson and Gia Bao Dong measured the VSWR, radiation pattern, and cross-polarization data, and J.C. Hanson produced the graphs.

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Data Availability Statement: Code and other resources available at https://github.com/918particle/Antenna-MEEP-project.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CEM	Computational Electromagnetism
LPDA	Log-periodic Dipole Array
MEEP	MIT Electromagnetic Equation Propagator
NEEC	Naval Engineering Education Consortium
NSWC Corona	Naval Surface Warfare Center, Corona Division
RF	Radio-frequency

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